I. PHONEBOOK MANAGEMENT SYSTEM.

The Phonebook management system is designed to assist users in conveniently and efficiently storing personal contact information.
The system encompasses the following core functional modules:

1. User Registration and Login

- ***User Registration:***

- New users can create an account by providing a username, email address, password, and password confirmation.
- The system validates the entered information and checks whether the username or email address already exists.
- If the information is valid, the system creates a new account and displays a successful registration message. Otherwise, an error message is displayed, and the user is required to correct the information.

- ***User Login:***

- Users enter their username and password to access the system.
- The system verifies the login credentials and checks the account status.
- If the credentials are correct and the account is active, the system grants access to the functionalities corresponding to the user's role.
- If the credentials are incorrect or the account is locked, the system displays an appropriate error message.

- ***User Logout:***

- Users can log out of the system when they no longer need to use it.
- After logout, the system terminates the current session and requires users to log in again to access contact management functionalities.

2. Viewing and Searching Contacts

- ***Viewing Contact Lists:***

- After logging in, users can view all contacts stored in their personal phonebook.
- Each contact may contain the following information: contact ID, full name, gender, occupation, address, date of birth, personal phone number, company phone number, personal email, company email, and notes.
- Users can select a contact to view its detailed information.
- The contact list can be sorted alphabetically by name or by creation date.

- ***Searching Contacts:***

- Users can search for contacts by entering keywords into the search field.
- The system returns a list of contacts whose names, phone numbers, or email addresses match the entered keywords.
- If no matching contacts are found, the system displays a notification indicating that no results are available.
- Users can only search for contacts belonging to their own phonebook.

3. Adding and Editing Contacts

- ***Adding New Contacts:***

- Users can select the Add Contact function to create a new contact.
- The system displays a form where users can enter the contact's full name, gender, occupation, address, date of birth, phone numbers, email addresses, and notes.

- The full name is mandatory. Each contact must have at least one phone number or email address.
- When users select Save, the system validates the entered information.
- If the information is valid, the system automatically generates a unique contact ID, stores the contact in the user's phonebook, and displays a success message.

- ***Editing Contact Information:***

- Users can select an existing contact to update its information.
- The system displays the current contact details.
- Users can modify the necessary fields and save the changes.
- The system validates the updated information, saves the changes, and displays a confirmation message.
- Users can only edit contacts that belong to their own phonebook.

4. Deleting Contacts

- Users can select one or multiple contacts from their phonebook for deletion.
- When users select the Delete function, the system displays a confirmation message.
- If users confirm the operation, the selected contacts are deleted from the phonebook.
- Any associations between the deleted contacts and their Tags or Favourite List are also removed.
- If users cancel the operation, the contact data remains unchanged.
- Users are not allowed to delete contacts belonging to other accounts.

5. Managing Contact Tags

- ***Creating Tags:***

- Users can create tags to categorize their contacts, such as Family, Friends, Work, or Customers.
- Each tag contains a tag ID, tag name, and description.
- When creating a tag, users enter its name and description. The system checks for duplicate tag names within the same account.

- ***Assigning Tags to Contacts:***

- Users can assign one or multiple tags to a contact.
- A single tag can be assigned to multiple contacts.
- For example, a contact may belong to both the Friends and Work categories.

- ***Editing and Deleting Tags:***

- Users can modify the name or description of an existing tag.
- Users can delete tags that are no longer needed.
- When a tag is deleted, the system removes the tag and its associated relationships without deleting the contacts assigned to it.

- **Filtering Contacts by Tags:**

- Users can select a tag to view all contacts associated with it.
- The system displays the matching contacts and allows users to view their detailed information.

6. Managing the Favourite List

- ***Adding Contacts to the Favourite List:***

- Users can mark a contact as a favourite.
- The system adds the selected contact to the user's Favourite List.
- A contact cannot appear more than once in the same Favourite List.

- ***Viewing Favourite Contacts:***

- Users can access their Favourite List to view contacts marked as favourites.

- The system displays the list of favourite contacts and allows users to view their detailed information.

- ***Removing Contacts from the Favourite List:***

- Users can remove a contact from their Favourite List.
- The system removes the contact from the Favourite List without deleting it from the main phonebook.
- If a contact is deleted from the main phonebook, it is automatically removed from the Favourite List.

7. Importing Contacts from Files

- Users can import multiple contacts into their phonebook using CSV files.
- Users select the Import Contacts function and upload a data file.
- The system validates the file format, data structure, and information of each contact.
- The system displays the validation results, including the number of valid contacts and any invalid records.
- Users confirm the import of valid contact records.
- The system saves the imported contacts to the user's phonebook and displays the number of successfully imported contacts.
- Imported contacts can be edited, deleted, assigned tags, or added to the Favourite List in the same way as manually created contacts.

8. User Account Management (Admin)

- ***Viewing User Accounts:***

- Admins can access the account management function to view all registered users.
- Each account displays basic information, including user ID, username, email address, and account status.
- Admins can search for accounts by username or email address.

- ***Locking User Accounts:***

- Admins can select a user account to lock.
- The system displays a confirmation message.
- After confirmation, the system changes the account status to **LOCKED** and invalidates any active sessions.
- Locked users cannot log in until their accounts are unlocked.
- Admins are not allowed to lock their own active accounts.

- ***Unlocking User Accounts:***

- Admins can select a locked account and perform the Unlock function.
- The system updates the account status to **ACTIVE.**
- Users can then log in and continue managing their personal contacts.

- ***Deleting User Accounts:***

- Admins can delete accounts that are no longer needed according to the system's account management policy.
- Before deletion, the system requires confirmation.
- When an account is deleted, the system handles its associated Contacts, Tags, and Favourite List according to the data retention and deletion policy.
- Admins cannot delete their own active administrator accounts.

9. System Statistics (Admin)

- Admins can view statistical information about the Phonebook Management System.
- The system provides the following statistics:
    - Total number of registered users.
    - Number of active user accounts.
    - Number of locked user accounts.
    - Total number of contacts stored in the system.
- The statistics are updated based on the current system data.
- Admins can only view aggregated statistics and are not automatically granted access to other users' private contact information.

II. NON-FUNCTIONAL REQUIREMENTS

2.1. Usability

- The system shall provide a simple, clear, and user-friendly interface.
- Users shall be able to easily navigate between the main functions of the system.
- The system shall display clear success and error messages after user actions.
- The system shall provide confirmation dialogs before performing important actions such as deleting contacts, deleting accounts, or locking accounts.
- The interface shall be responsive and support desktop, tablet, and mobile devices.

2.2. Reliability

- The system shall operate stably during normal usage.
- The system shall ensure that user data is stored accurately and consistently.
- The system shall prevent data loss when adding, editing, or deleting contacts.
- The system shall handle errors without affecting previously saved data.
- The system shall support data backup and recovery to minimize data loss.

2.3. Performance

- The system shall support at least 20 concurrent users under normal operating conditions.
- Common operations, such as logging in, viewing contacts, searching, adding, editing, and deleting contacts, shall respond within 2 seconds under normal load.
- The system shall provide fast search results even when users have a large number of contacts.
- The system shall process CSV files efficiently during contact imports.
- The system shall generate system statistics within a reasonable time.

2.4. Supportability and Maintainability

- The system shall be designed with a modular architecture to simplify maintenance and future development.
- The source code shall follow consistent naming conventions and coding standards.

- The system shall allow developers to modify or add features without significantly affecting existing functions.
- The system shall maintain error logs to support troubleshooting.
- The database structure shall be designed to support efficient maintenance and updates.

2.5. Security

- The system shall store passwords using secure password-hashing algorithms.
- The system shall require authentication before users can access protected functions.
- The system shall implement Role-Based Access Control (RBAC) to distinguish between User and Admin permissions.
- Users shall only be allowed to view and manage their own contacts, tags, and favourite lists.
- Only Admin accounts shall be allowed to access account management and system statistics.
- The system shall protect against common security vulnerabilities, including SQL Injection and Cross-Site Scripting (XSS).
- The system shall automatically invalidate active sessions when an account is locked.
- Important administrative actions shall be recorded in system logs.

2.6. Data Integrity

- Each user account, contact, and tag shall have a unique identifier.
- Usernames and email addresses shall be unique across the system.
- The system shall ensure that each user's data remains isolated from other users.
- The system shall maintain referential integrity between contacts, tags, and favourite lists.
- Deleting a contact shall also remove its related tag associations and favourite entries.
- Deleting a tag shall not delete the associated contacts.
- The favourite list shall not contain duplicate contact entries.
- The system shall validate input data before saving it to the database.

2.7. Compatibility

- The system shall support commonly used web browsers, including Google Chrome, Microsoft Edge, and Mozilla Firefox.
- The system shall be accessible on devices running Windows, macOS, Android, and iOS through supported web browsers.
- The system shall support CSV files following the required import format.
- The system shall maintain consistent functionality across supported browsers and devices.

2.8. Scalability

- The system shall support an increasing number of user accounts and contacts without requiring major architectural changes.
- The database shall allow additional contact fields to be added when necessary.
- The system architecture shall support the future addition of new features.
- The system shall be designed to allow increased processing capacity when the number of concurrent users grows.